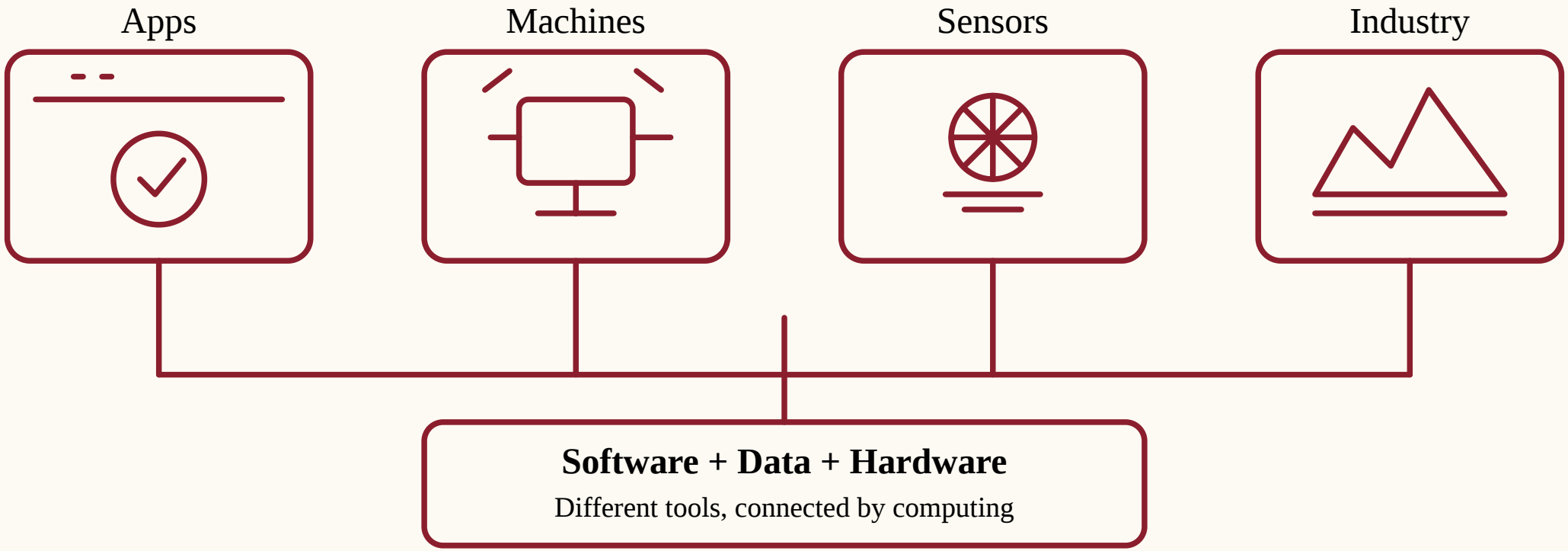
# WHAT

## Can You Build With Computer Science?

FROM CODE AND COMPUTERS TO REAL-WORLD SYSTEMS

J. L. Pach

# COMPUTER SCIENCE IS EVERYWHERE




**Computing connects the digital world with the physical world.**

# DIFFERENT PATHS, SHARED IDEAS

## Computer Science

- Algorithms and problem-solving
- Programming and data
- Artificial intelligence

## Software Engineering

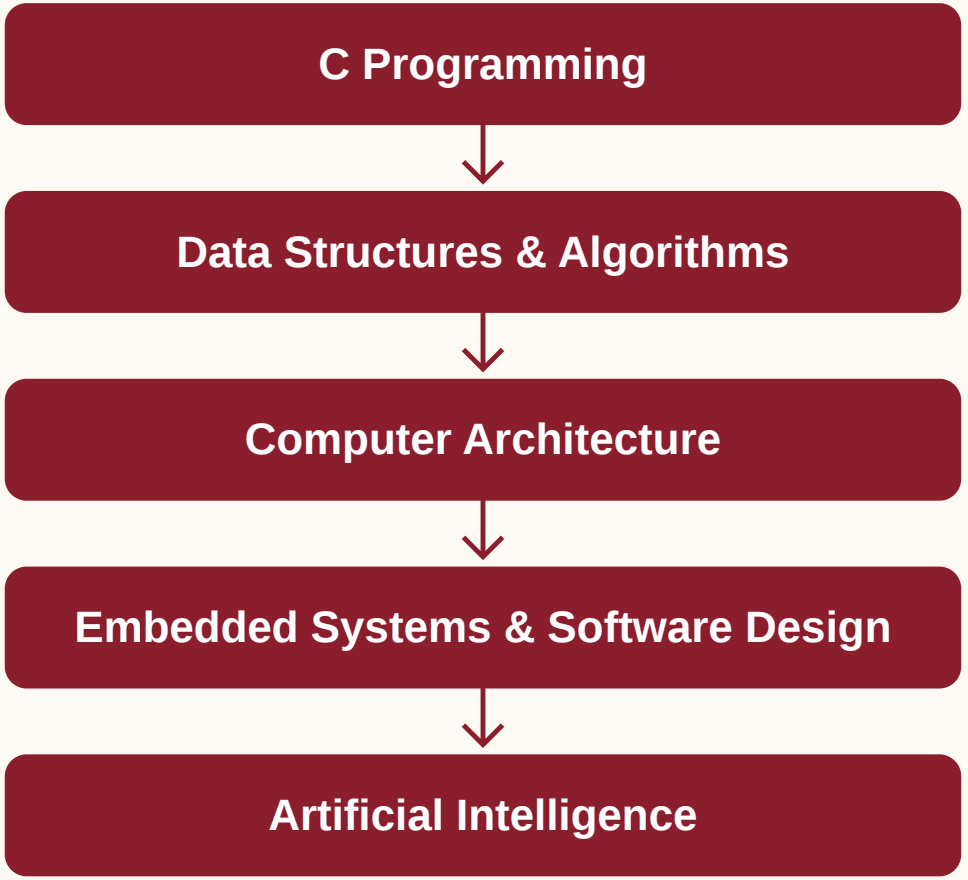
- Software design
- Testing and debugging
- Building systems in teams

## Cybersecurity & Networking

- Networks and communication
- System security
- Protecting information

**These fields overlap — and that is a strength.**

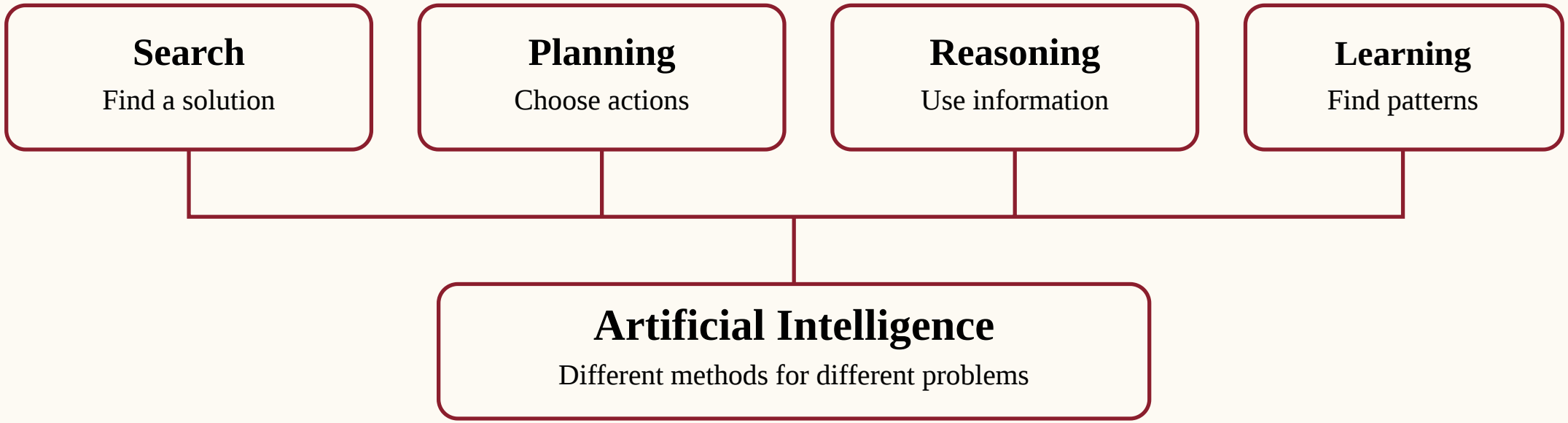
# FROM BITS TO INTELLIGENT SYSTEMS




One example of how computing topics connect

# AI IS MORE THAN A CHATBOT

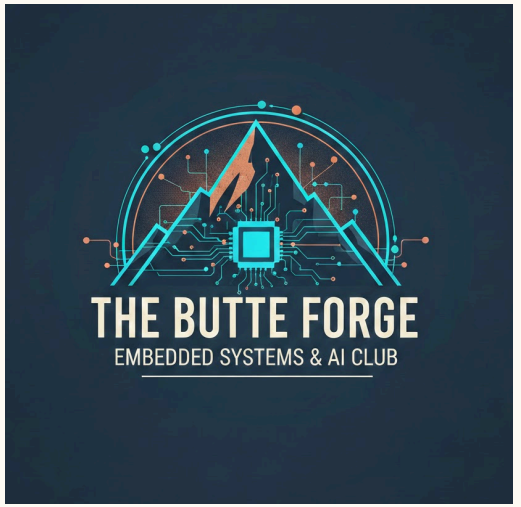
**Using AI is one skill. Understanding AI is another.**



# THE BUTTE FORGE

## Embedded Systems & AI

- Student-led, project-driven work
- Electronics and embedded devices
- Sensors and communication
- Applied AI
- Teamwork and hands-on learning




**Learn by building. Learn by testing. Learn by working together.**

# THE DUAL-TRACK MODEL - SENTINEL

**Track 1**: *Project SENTINEL*

- **Goal**:
  A multi-stage, high-end engineering project (Sonar/Lidar/AI).
- **Role**:
  J.L. Pach as Chief Architect.
- **Purpose**:
  Building a credible engineering portfolio and showcasing the level of engineering expected at Montana Tech.

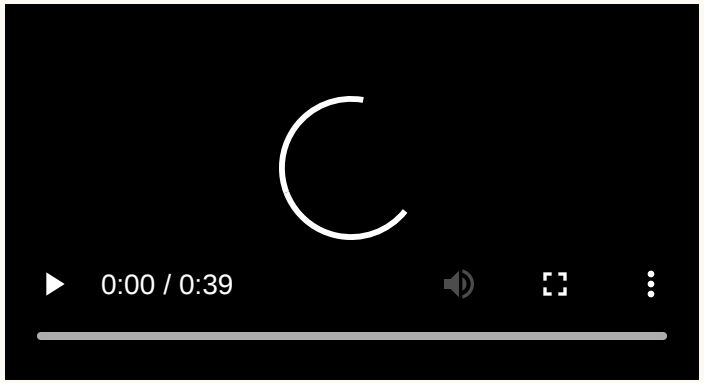


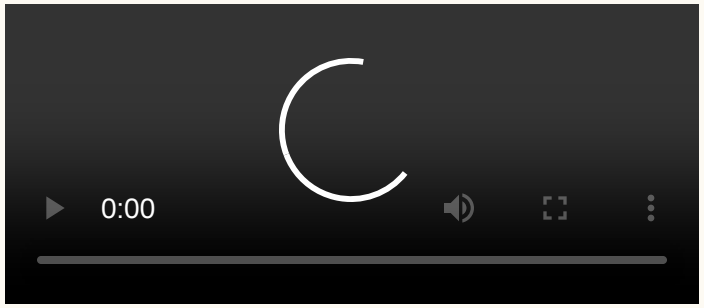
# THE DUAL-TRACK MODEL - SANDBOX

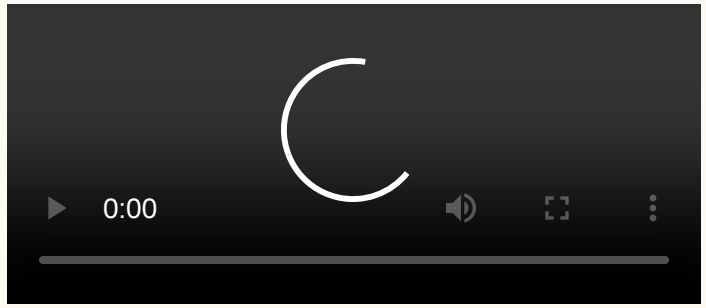
- **Track 2**: Project X (The Sandbox)
  - **Goal**:
    Student-led innovation (e.g., Retro Gaming, Smart Mirror, Automation).
  - **Role**:
    J. L. Pach as Consultant & Mentor.
  - **Purpose**:
    Creativity, ownership, and engineering for fun.



# SENTINEL PROJECT




- HC-SR04 sonar sensor to detect distance as the robot's `eyes` through emitting signals from Raspberry Pi
- Audio sampling and BPM extraction from live audio using audio buffers and Fast Fourier Transform to convert from frequency to BPM
- Python scripts controlling individual servos on the robot with logic to `dance`
- Scripting output using TCP/IP socket to Webots script for virtual simulation

# SATURDAY SAFARI ZONE





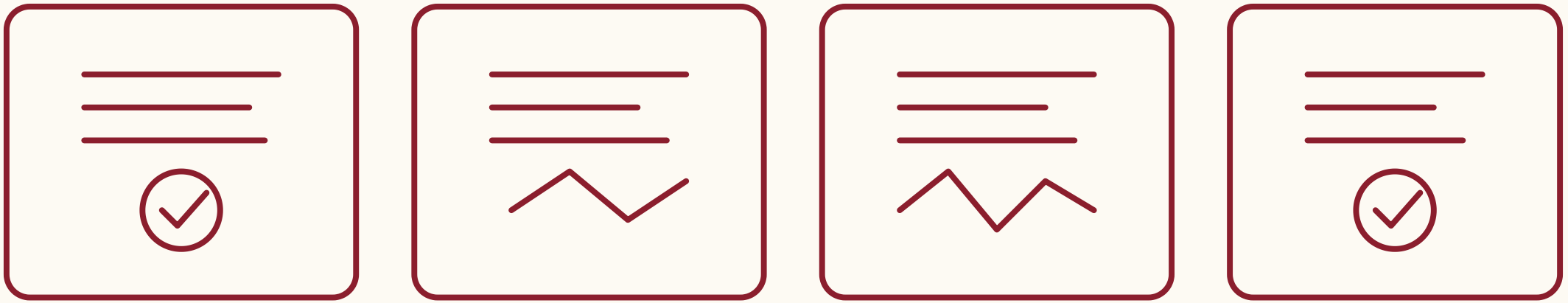








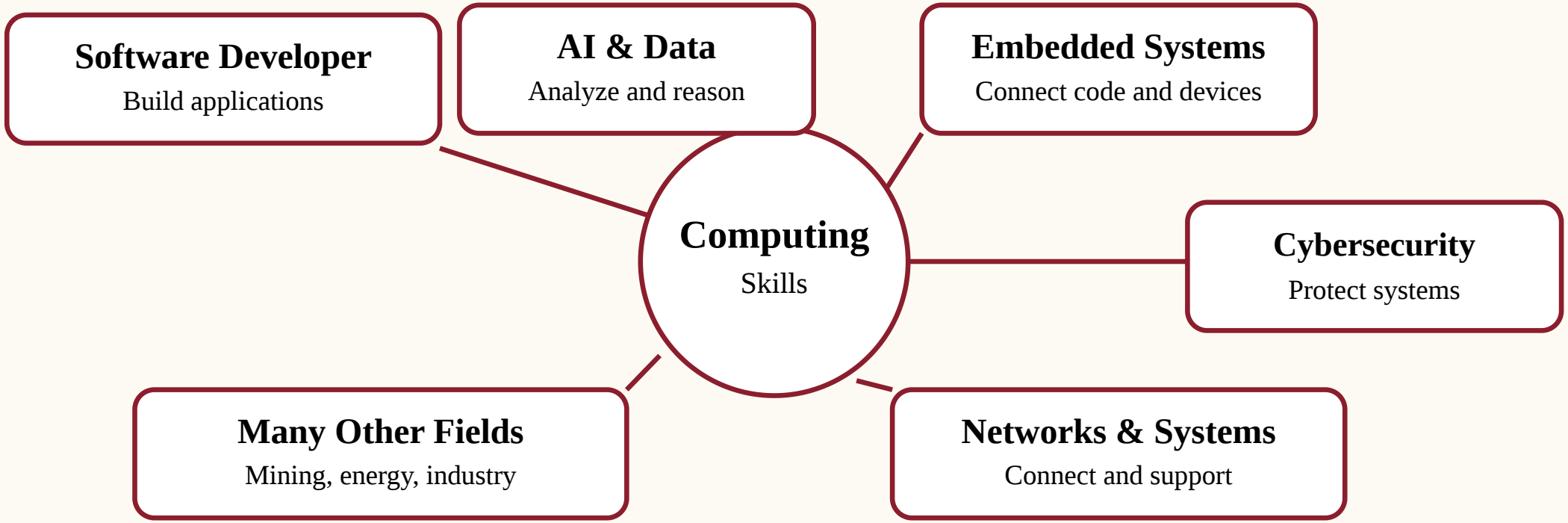
# WHAT YOU ACTUALLY LEARN



**Code** **Algorithms** **Testing** **Teamwork**

Build it. Test it. Understand it. Improve it.

# WHERE CAN THESE SKILLS TAKE YOU?




**Computing skills can support many kinds of engineering work.**

# YOU DON'T NEED TO KNOW EVERYTHING YET

- Try different areas.
- Build small projects.
- Ask questions.
- Learn from mistakes.
- Find people who share your interests.

**START CURIOUS. GET BETTER BY DOING.**

# START WITH A PROBLEM. BUILD A SOLUTION.

- Learn the fundamentals.
- Try projects.
- Find your direction.

# THANK

**you!**

# QUESTIONS?